

EE 115 Homework 4

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Problem 1

Let $m(t) = 2 \operatorname{sinc}(3t)$ where t is in milliseconds. With the engineering definition $\operatorname{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, the Fourier transform is

$$M(f) = \frac{2}{3} \operatorname{rect}\left(\frac{f}{3}\right),$$

so the baseband bandwidth is $B = 1.5$ kHz and $f_c = 4B = 6$ kHz. The input to the nonlinear device is $x_{\text{in}}(t) = m(t) + \cos(2\pi f_c t)$, and the device output is

$$x_{\text{out}}(t) = \alpha x_{\text{in}}^2(t) - \beta x_{\text{in}}(t).$$

- (a) The spectrum of $x_{\text{out}}(t)$ contains (i) a low-pass term from $\alpha m^2(t)$, (ii) shifted replicas about $\pm f_c$ from $2\alpha m(t) \cos(2\pi f_c t)$, (iii) spectral lines at $\{0, \pm f_c, \pm 2f_c\}$ from the carrier harmonics, and (iv) the attenuated baseband $-\beta m(t)$ plus the carrier $-\beta \cos(2\pi f_c t)$. Figure 1 overlays these pieces.

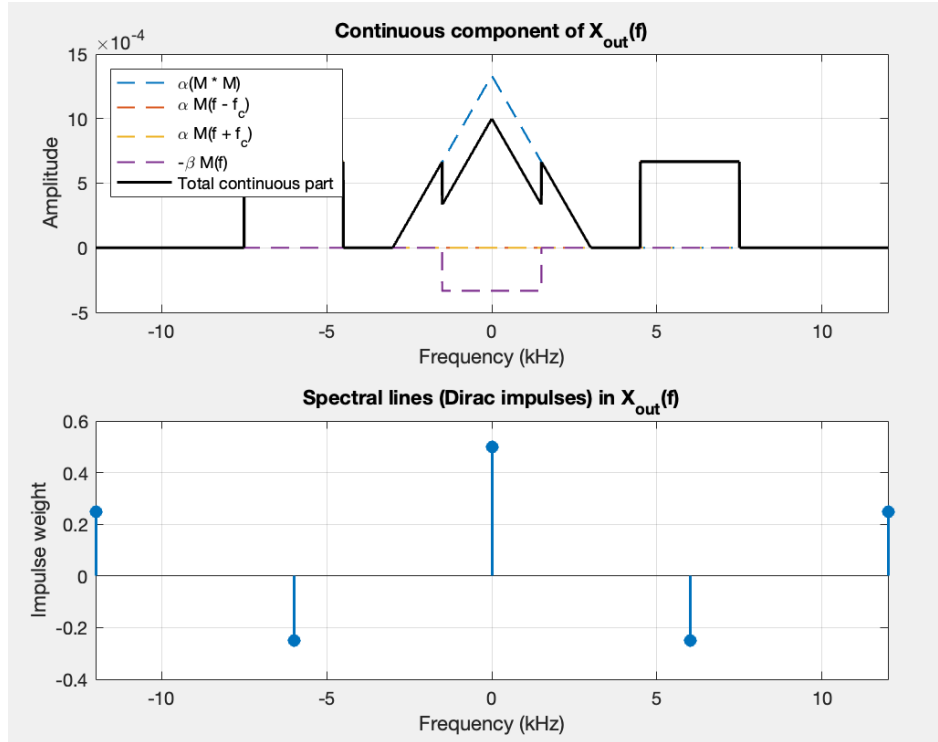


Figure 1: Spectral sketch for $X_{\text{out}}(f)$ highlighting the continuous components and the Dirac impulses.

- (b) The bandpass filter must extract only the components around $\pm f_c$ while rejecting the baseband ($|f| \leq B$) and the second-harmonic clusters near $\pm 2f_c$. A suitable ideal response is $H(f) = 1$ for $f_c - B \leq |f| \leq f_c + B$ (zero elsewhere) with $f_c = 6$ kHz and $B = 1.5$ kHz. The filtered output is $u_{\text{AM}}(t) = (-\beta + 2\alpha m(t)) \cos(2\pi f_c t)$, a conventional AM waveform consisting of a carrier plus a scaled copy of the message.
- (c) For envelope detection without distortion, the instantaneous amplitude must remain nonnegative: $|2\alpha m(t)| \leq \beta$ for all t . Since $|m(t)| \leq 2$, the positive parameters must satisfy $\boxed{\beta \geq 4\alpha}$. The detected envelope is $\beta - 2\alpha m(t)$; removing the DC component β (with a blocker) and scaling by $\frac{1}{2\alpha}$ yields $-m(t)$.

Problem 2

The message spectrum is the triangular function

$$M(f) = \begin{cases} 10 + f, & -10 \leq f \leq 0, \\ 10 - f, & 0 \leq f \leq 10, \\ 0, & \text{otherwise,} \end{cases}$$

and the DSB-SC signal is $u_1(t) = m(t) \cos(200\pi t)$. The bandpass filter $H(f)$ keeps only the slices $|f - 105| \leq 5$ and $|f + 105| \leq 5$.

- (a) The filter suppresses the lower sideband (frequencies in $[90, 100]$ Hz and $[-100, -90]$ Hz) and retains the upper sideband near ± 105 Hz. The resulting spectrum is plotted in Figure 2.

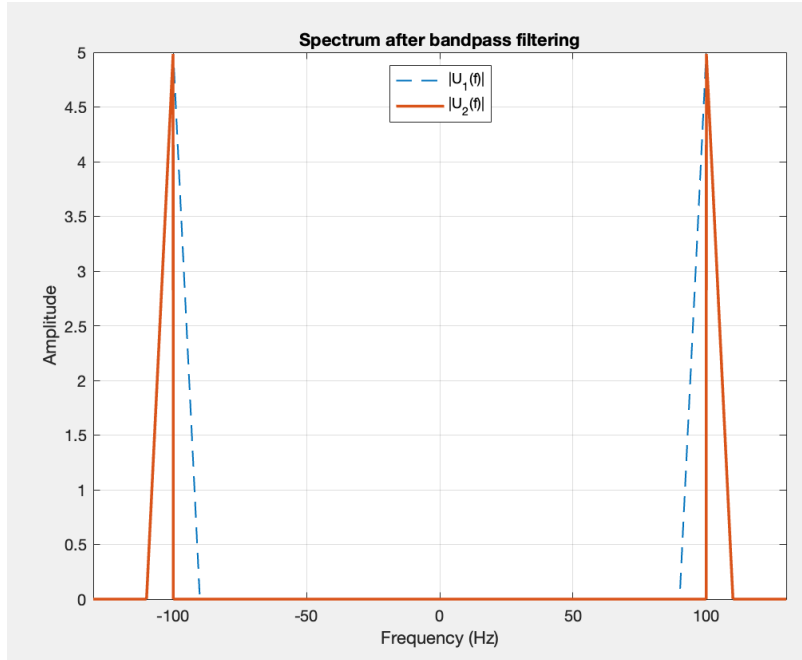


Figure 2: Magnitude spectra $|U_1(f)|$ and $|U_2(f)|$; the filter preserves only the upper sideband around ± 105 Hz.

- (b) Let $f_c = 100$ Hz. Removing the lower sideband leaves the complex envelope

$$s(t) = \int_0^{10} (10 - \nu) e^{j2\pi\nu t} d\nu = \frac{1 - e^{j20\pi t}}{(2\pi)^2} + j \frac{5}{\pi t},$$

with finite limits $s(0) = 50$ and $s'(0) = 0$. Therefore

$$u_2(t) = u_c(t) \cos(2\pi f_c t) - u_s(t) \sin(2\pi f_c t), \quad u_c(t) = \frac{1 - \cos(20\pi t)}{4\pi^2 t^2}, \quad u_s(t) = \frac{5}{\pi t} - \frac{\sin(20\pi t)}{4\pi^2 t^2},$$

with $u_c(0) = 50$ and $u_s(0) = 0$ understood by continuity.

- (c) Yes. Because $H(f)$ passes only the upper sideband of the original DSB-SC spectrum, the complex envelope $s(t) = u_c(t) + ju_s(t)$ contains only nonnegative baseband frequencies ($\nu \geq 0$). Consequently, $u_s(t)$ is the Hilbert transform of $u_c(t)$, which is characteristic of single-sideband AM signals obtained from analytic (positive-frequency) spectra.